

C.V RAMAN GLOBAL UNIVERSITY
MID-SEMESTER EXAMINATION 2025-26 (ODD)

Prog: B.Tech	Subject :- Basic Electrical Engineering	Time: 1 Hour
Branch: EE/EEE	Sub. Code: EE100	Max marks: 20
Semester: 1 st	Group-1,3,5,7,9	

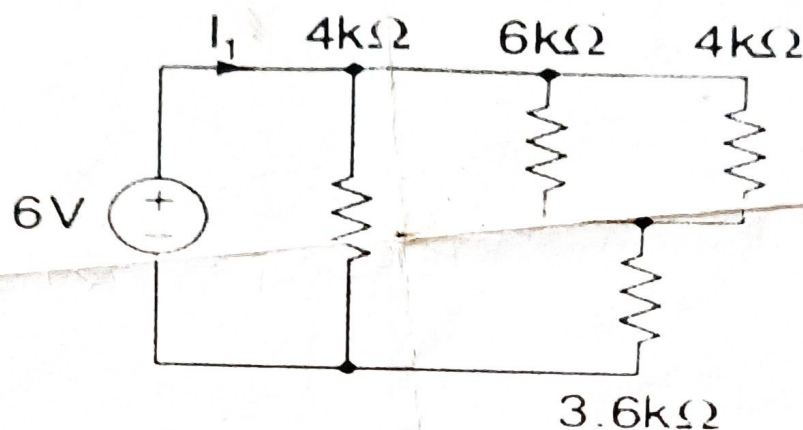
The figures in the right-hand margin indicate marks.

PART A

Answer ALL questions:

- | | | |
|----|--|--------------|
| a) | What is the Superposition Theorem? | [5x2] |
| b) | What is active power and what is power factor? | 2 |
| c) | What will be the power factor if, in an AC circuit, the voltage is $(8+6j)$ V and the current is $(7+8j)$ A? | 2 |
| d) | If the voltage in an ac circuit, $8\angle 60^\circ$ V and the impedance is $(9+2j) \Omega$, what will be the current magnitude? | 2 |

e)



Find the values of I_1 and current through $4k\Omega$ resistor.

PART B

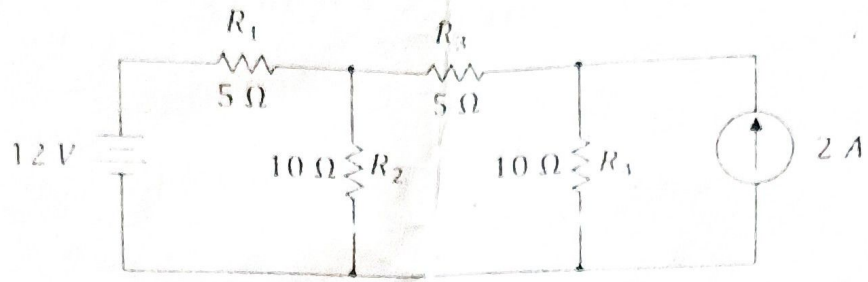
ANSWER ALL

- | | | |
|----|--|------------|
| a) | An alternating voltage of $v = 180 \sin(\omega t + 65^\circ)$ V and $i = 20 \sin(\omega t - 20^\circ)$ A is applied to a circuit consisting of a coil. What is the maximum voltage, rms voltage, rms current, real power, reactive power, power factor and impedance of the circuit. | [5] |
|----|--|------------|

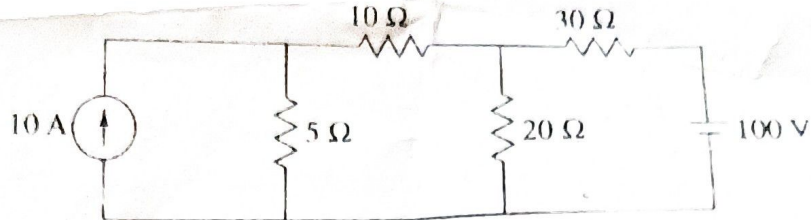
or

A resistor of 10Ω , and a capacitor of 50 mF are connected in series. A supply voltage 100 V , 50 Hz is connected across the series combination. Calculate the following: (i) impedance, (ii) maximum voltage, (iii) rms voltage, (iv) maximum current drawn by the circuit, (v) phase difference and power factor, and (vi) active and reactive power consumed by the circuit

b) Find the current through R_3 resistor by using any method.



or



Find the current through 10Ω using Thevenin's theorem

Course Outcome Assessment Scheme:

CO-1= 45%	Q.1 (a, e); Q.2 (b)	=2+2+5
CO2= 55%	Q.1 (b, c, d); Q.2(a)	=2+2+2+5
Bloom's Level	Q.1 (a, b)	L-2
	Q.1 (c, d,e): Q.2. (a),(b)	L-3

Total number of printed pages – 3

B.Tech

Subject Code: EE100

✓ 1st SEMESTER REGULAR EXAMINATION –2025-26

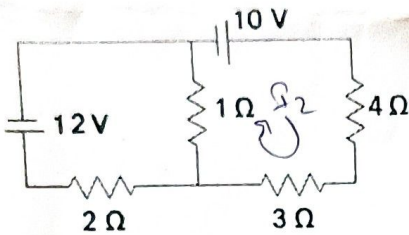
SUBJECT NAME: BASIC ELECTRICAL ENGINEERING

(GROUP-1, 3, 5, 7, 9)

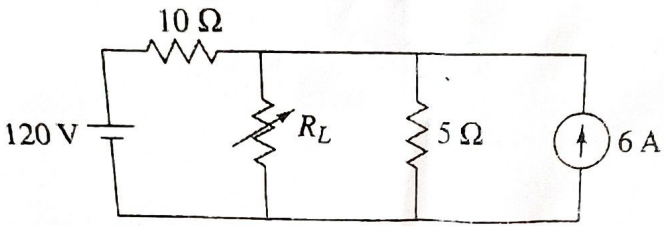
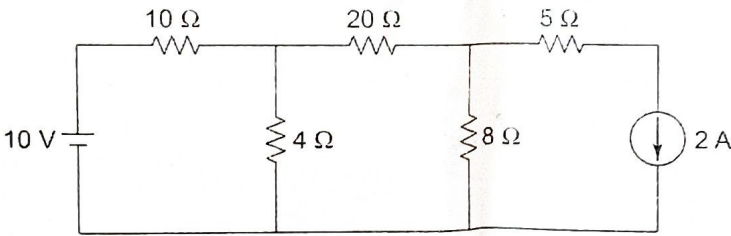
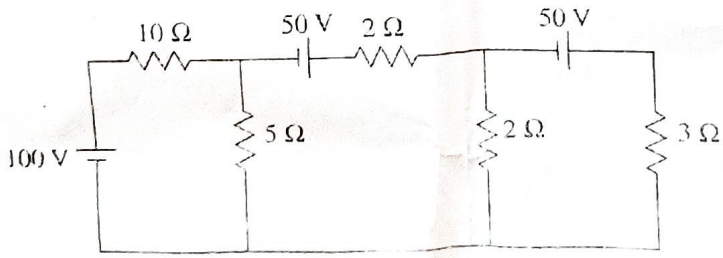
Time – 3 Hours

Full Marks – 100

Answer all Questions from Part – A and Part – B

Q1.	Part-A (ANSWER ALL QUESTIONS) ,	CO	Marks	BL
(a)	State maximum power transfer theorem.	1	2	2
(b)	What are the limitations of superposition Theorem?	1	2	2
(c)	 Find the current through 4 ohm 0.78	1	2	3
(d)	Find 440V, 50 Hz supply is connected with a Star connected three-phase load then what will be the line voltage.	2	2	3
(e)	Draw the impedance triangle for the R-L-C series circuit and write the expression for the power factor from it	3	2	2
(f)	If the voltage in an AC circuit, $10\angle 60^\circ$ V and the impedance is $(6+8j)\ \Omega$, what will be the current magnitude? (1)	2	2	3
(g)	What are the application of BLDC motor?	2	2	2
(h)	For a magnetic circuit flux is 10 wb and reluctance is 5 AT/wb. What will be the mmf? 50	2	2	3
(i)	Define Magnetomotive Force (MMF) and Reluctance in a magnetic circuit. Explain the relationship between MMF, Reluctance.	3	2	3

(j)	For a single-phase transformer find the secondary induce emf, if the number of turns in the secondary is 60, the maximum flux in the core is 20wb and the supply voltage is 220V at 50Hz	4	2
(k)	Differentiate between core type and shell type Transformer?	4	2
(l)	Why are transformers rated in kVA?	4	2
(m)	Analyse the advantages of PMMC type instrument.	5	2
(n)	Differentiate between MI- repulsion type and MI- attraction type measuring instruments.	5	2
(o)	Why we need earthing? What are the different types of it?	5	2

Part-B		CO	Marks	
Q2.	Answer any two:			
(a)	Find the value of ' R_L ' in Fig. such that maximum power will be delivered to load 	1	7	
(b)	Calculate the current in 20ohm in the circuit shown in Fig. using nodal analysis 	1	7	3
(c)	 Find the current in 2ohm using mesh analysis 15.81A 0.51A 2 20.4 10.4 2 5.2	1	7	3

Q3.	Answer any two:			
(a)	Three coils each having a resistance of 20Ω and reactance of 10Ω are connected in Star to a 440V, three-phase, 50 Hz supply. Calculate (i) line current (ii) phase voltage (iii) power factor (iv) power consumed by the load (v) reactive power	2	7	
(b)	An alternating voltage of $v = 180 \sin(\omega t + 85^\circ)$ and $i = 60 \sin(\omega t - 25^\circ)$ is applied to a circuit consisting of a coil. What is the maximum voltage, rms voltage, rms current, real power, reactive power, power factor and impedance of the circuit.	2	7	3
(c)	Two impedances $Z_1 = 20 + j10$, and $Z_2 = 30 - j10$ are connected in series across a 220V, 50Hz ac supply. Calculate the (i) impedance, (ii) current, (iii) power factor of the total circuit, and (iv) Power consumption of the circuit.	2	7	3
Q4.	Answer any two:			
(a)	How are DC motors constructed and explain the principle of operation?	3	7	3
(b)	Explain nature of B-H curve for magnetic material with a neat diagram.	3	7	3
(c)	What is ohms law for magnetic circuit An air gap 1.1mm long and 40 sq cm in cross-section exists in a magnetic circuit. Determine i. Reluctance of air gap. ii) MMF required to create a flux of 10×10^4 wb in the air gap	3	7	3
Q5.	Answer the followings:			
(a)	Derive the EMF equation for the transformer. A 230/120 V, 40 kVA Transformer has 60 turns in the secondary. Calculate the primary number of turns & primary and secondary full load currents neglecting the losses.	4	7	3
(b)	How three-phase induction motor works? Explain the production of rotating magnetic field in three-phase induction motor	4	7	2
Q6.	Answer the followings:			
(a)	How does the MI-type (repulsion type) measuring instrument work? Differentiate it from PMMC type instrument.	5	7	2
(b)	Explain the concept of power transmission and power distribution. Draw and label a block diagram showing how electrical power is transmitted from generating stations to consumers through distribution system.	5	7	2